Geospatial Analysis and Visualisation at ADA

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Presentation Overview

1. Why geospatial visualisation
2. About the GIS visualisation tool
3. Components of the GIS tool
4. Map options
5. Types of variables: nominal, ordinal, interval
4. Map types by variable type
   a) Percentages, modes and histograms
   b) Histograms, radars and spie charts
   c) Colour schemes
   d) Chloropleth and dorling
5. Data export
6. Future directions

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ASSDA or ADA?

- From July 2011, ASSDA will be changing names to ADA - The Australian Data Archive, with a new website:
  - http://www.ada.edu.au

ADA AUSTRALIAN DATA ARCHIVE

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1. Why geospatial visualisation?

- Search
- Exploration
- Validation
- Cross disciplinary relationships (Mashups)

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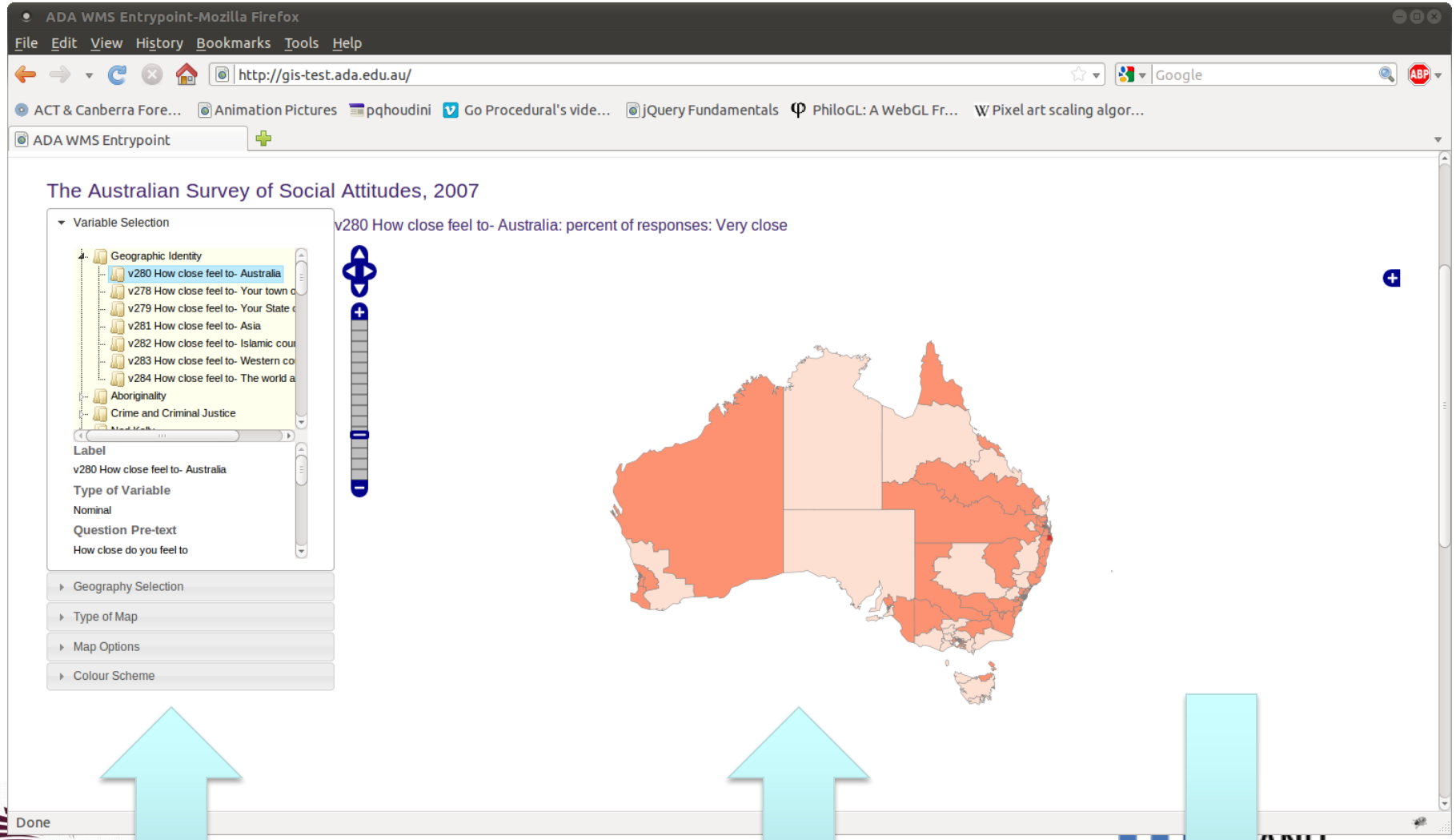
The ADA GIS Visualisation tool



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Open technologies

- Web mapping service (WMS)
- OpenLayers: web map rendering
- Postgresql with PostGIS extensions: database
- jQuery: user interface
- Ruby on Rails: CMS
- DDI
- Python

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Components of the GIS Visualisation tool

- Left-hand menu
- Map interface
- Legend and scale
- ADA look and feel

http://gis-test.ada.edu.au/ (Dev server only – password protected :-)

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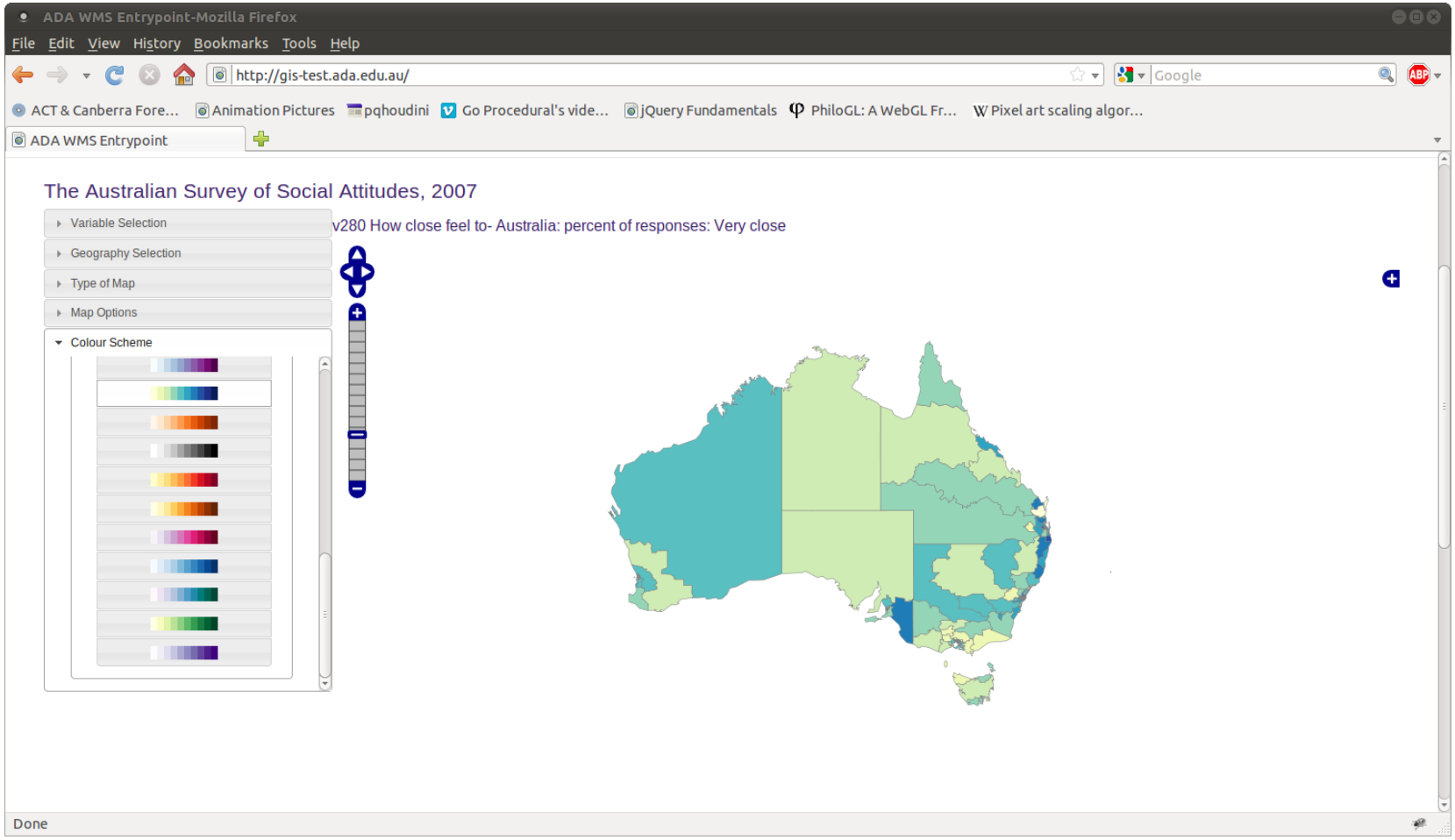
Changing colour schemes



Colours from Colorbrewer: http://colorbrewer2.org/

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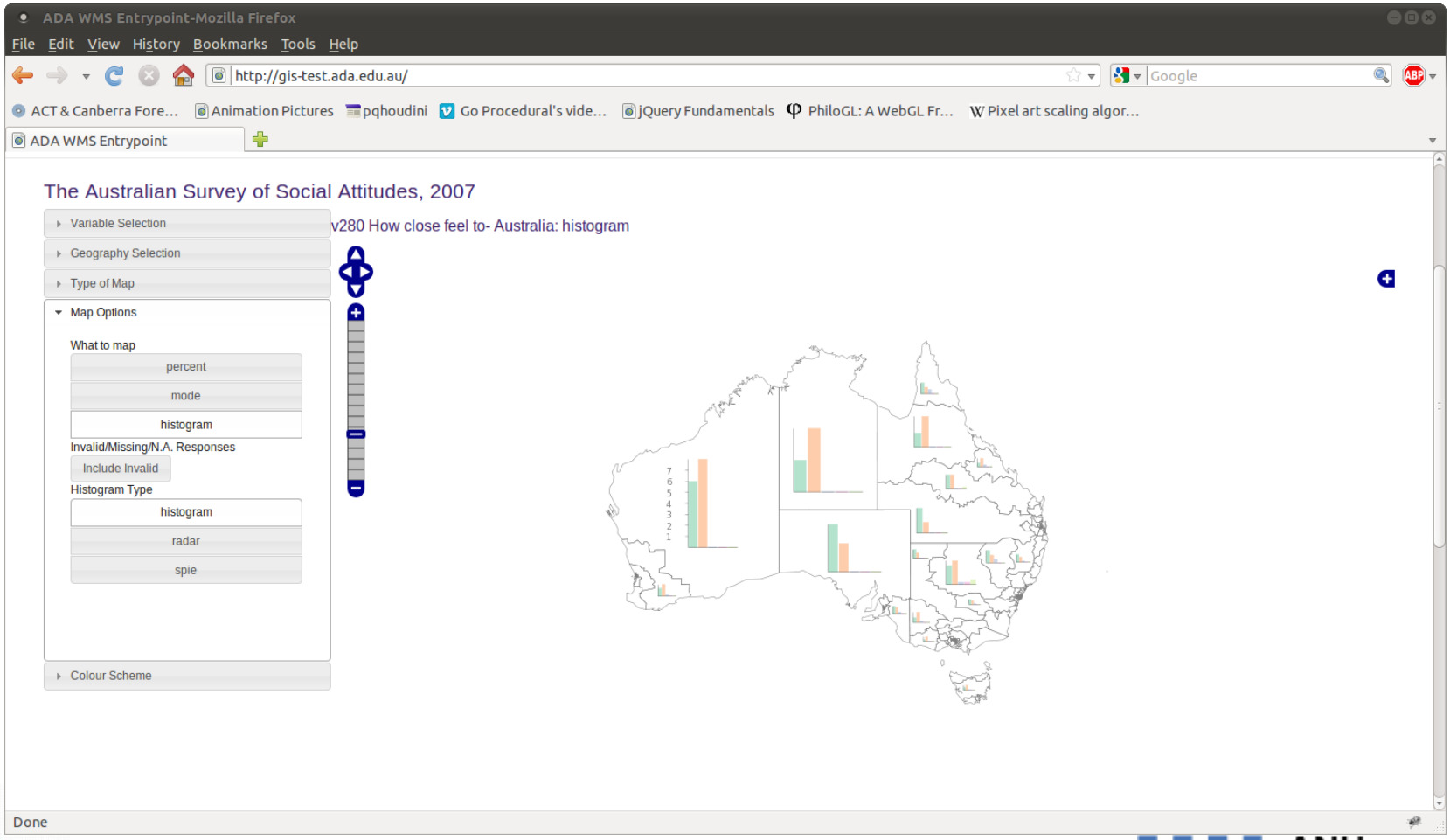
Map options



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Options for different variable types

Dataset: Australian Survey of Social Attitudes 2007 (includes ISSP questions)

- Interval: v559 Year born
- Ordinal: v533 Self-rated position in society
- Nominal: v66 Urgent environmental concern-Climate Change

Map options vary depending on variable type

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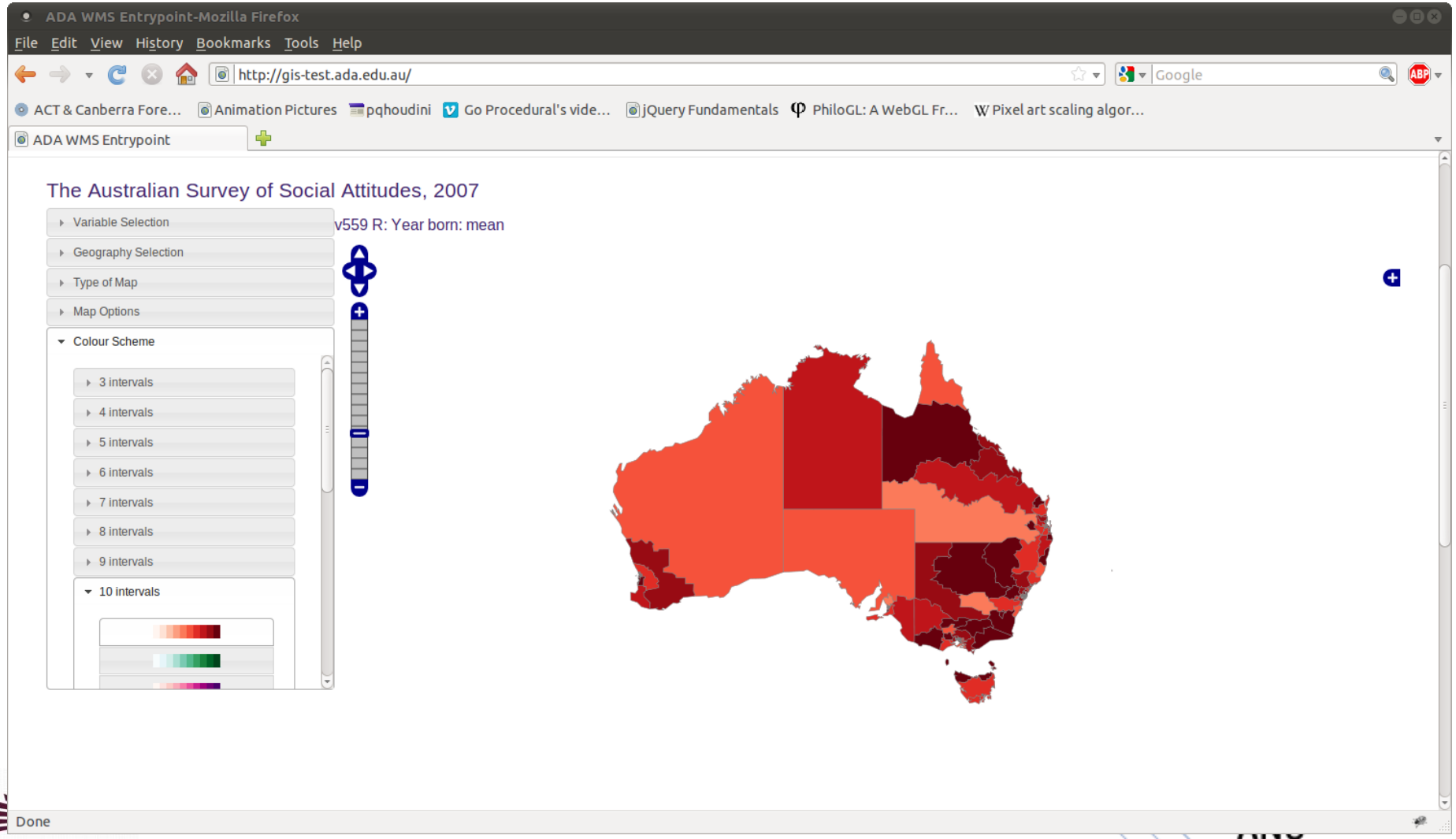
Interval variables



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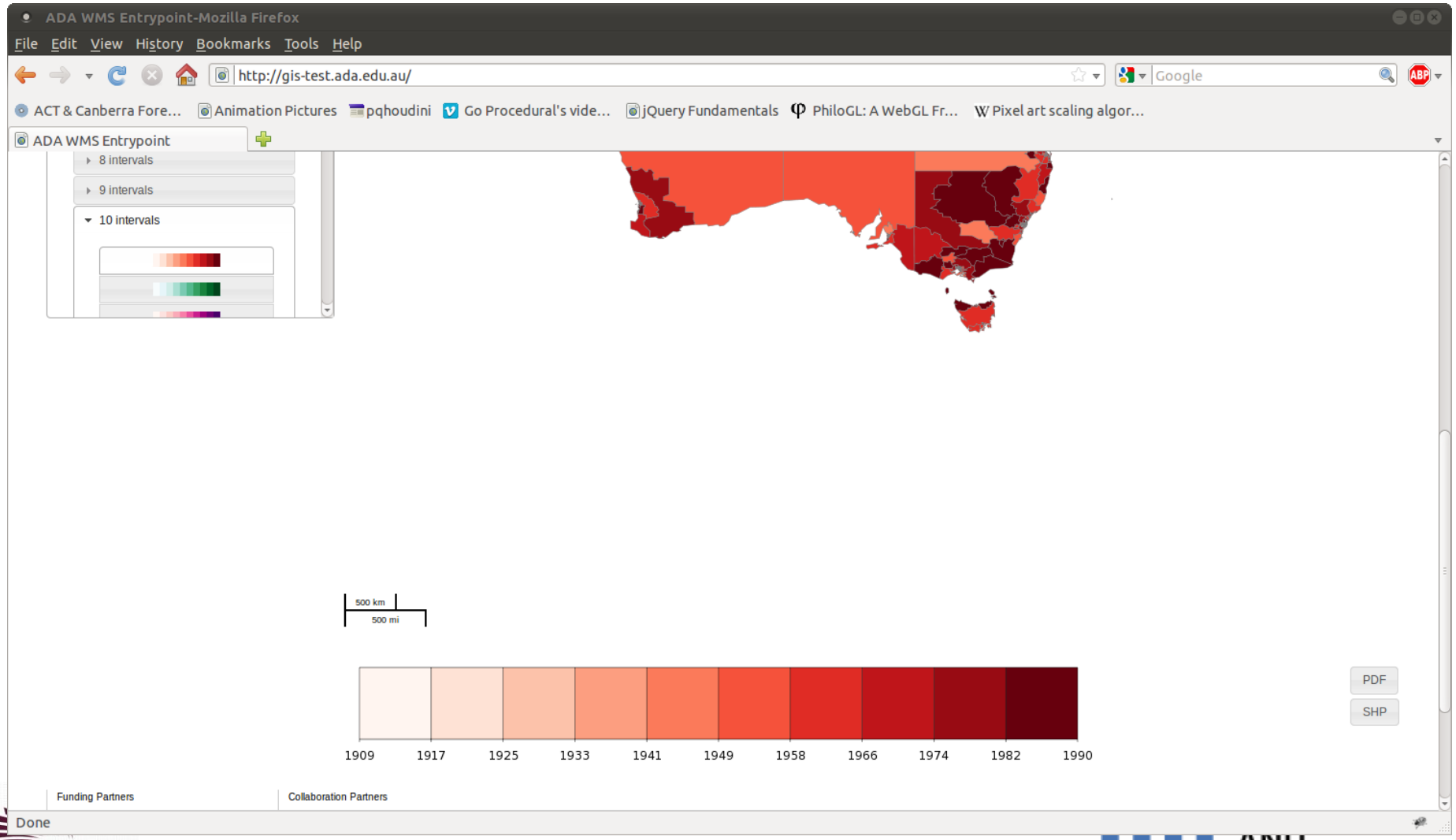
Interval variables



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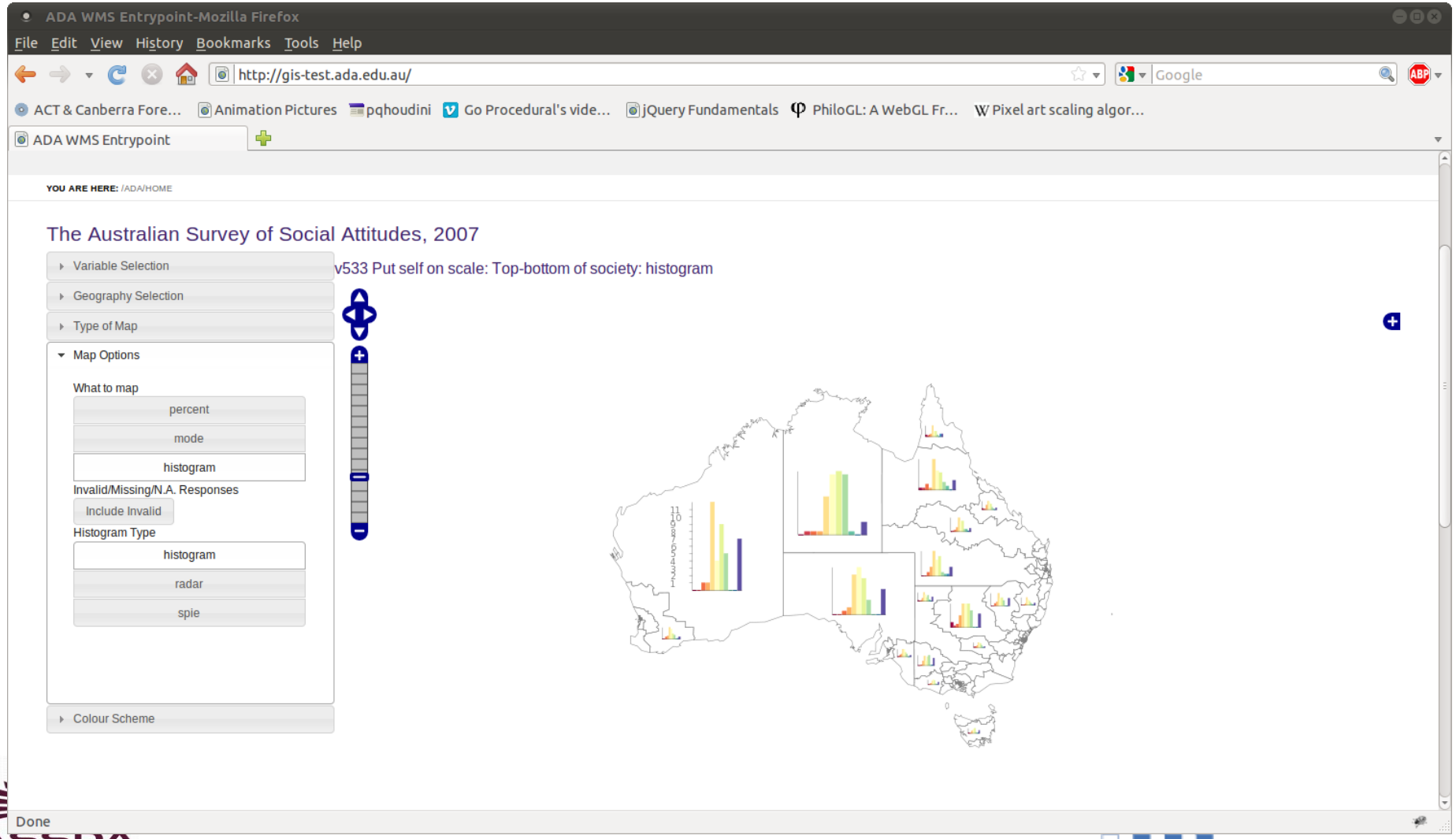
Ordinal variables - histograms



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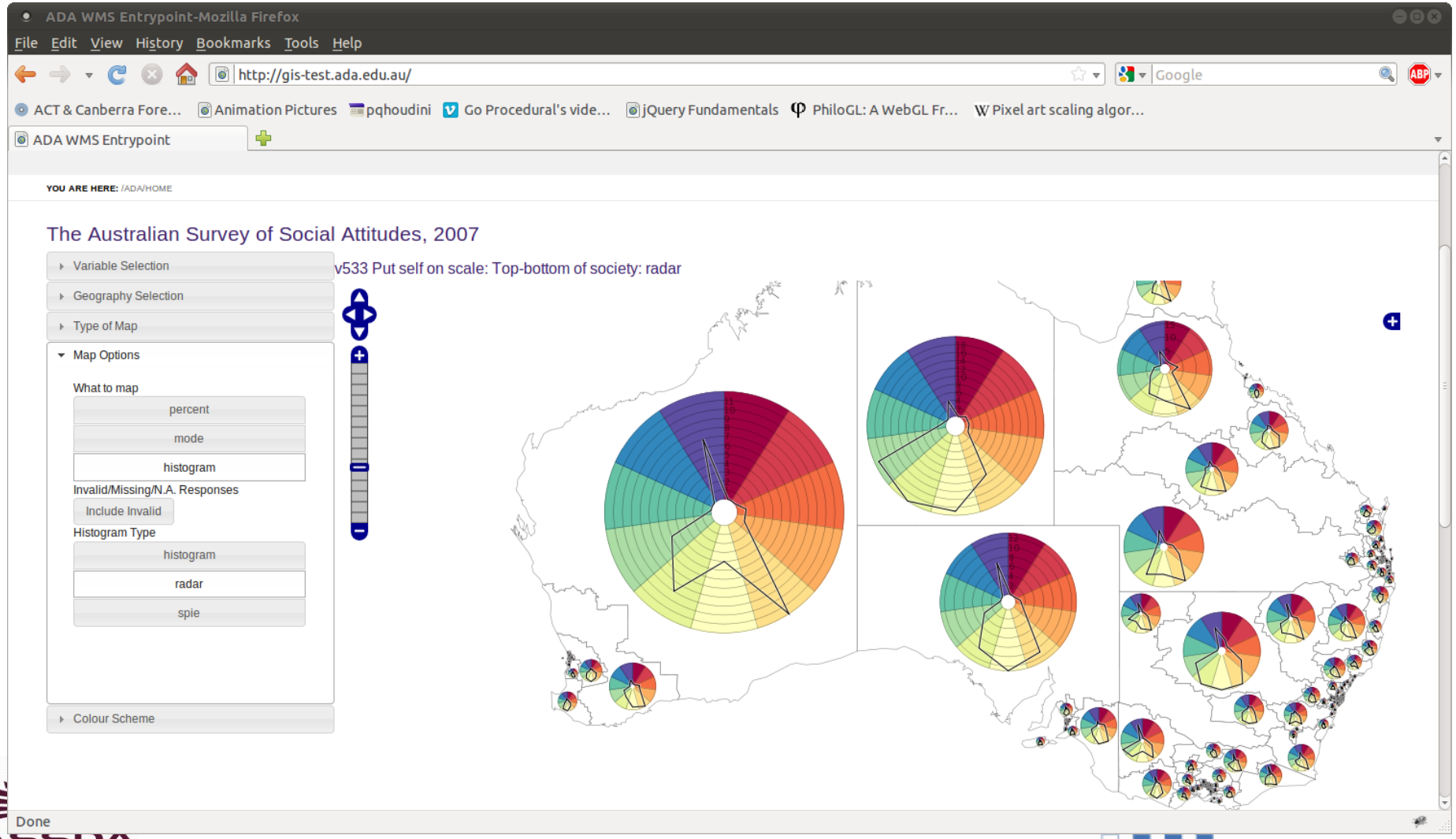
Ordinal variables – radar plots



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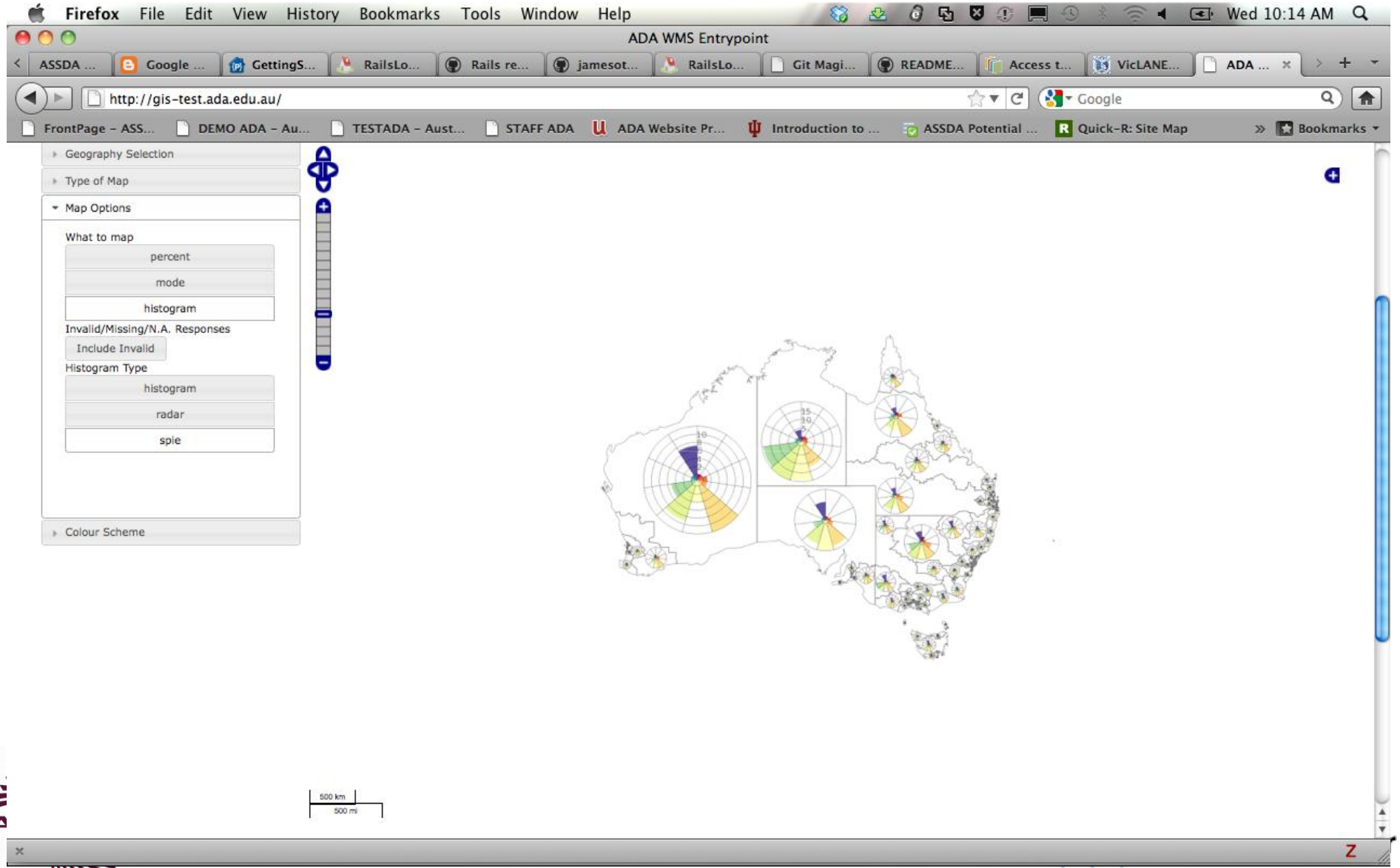
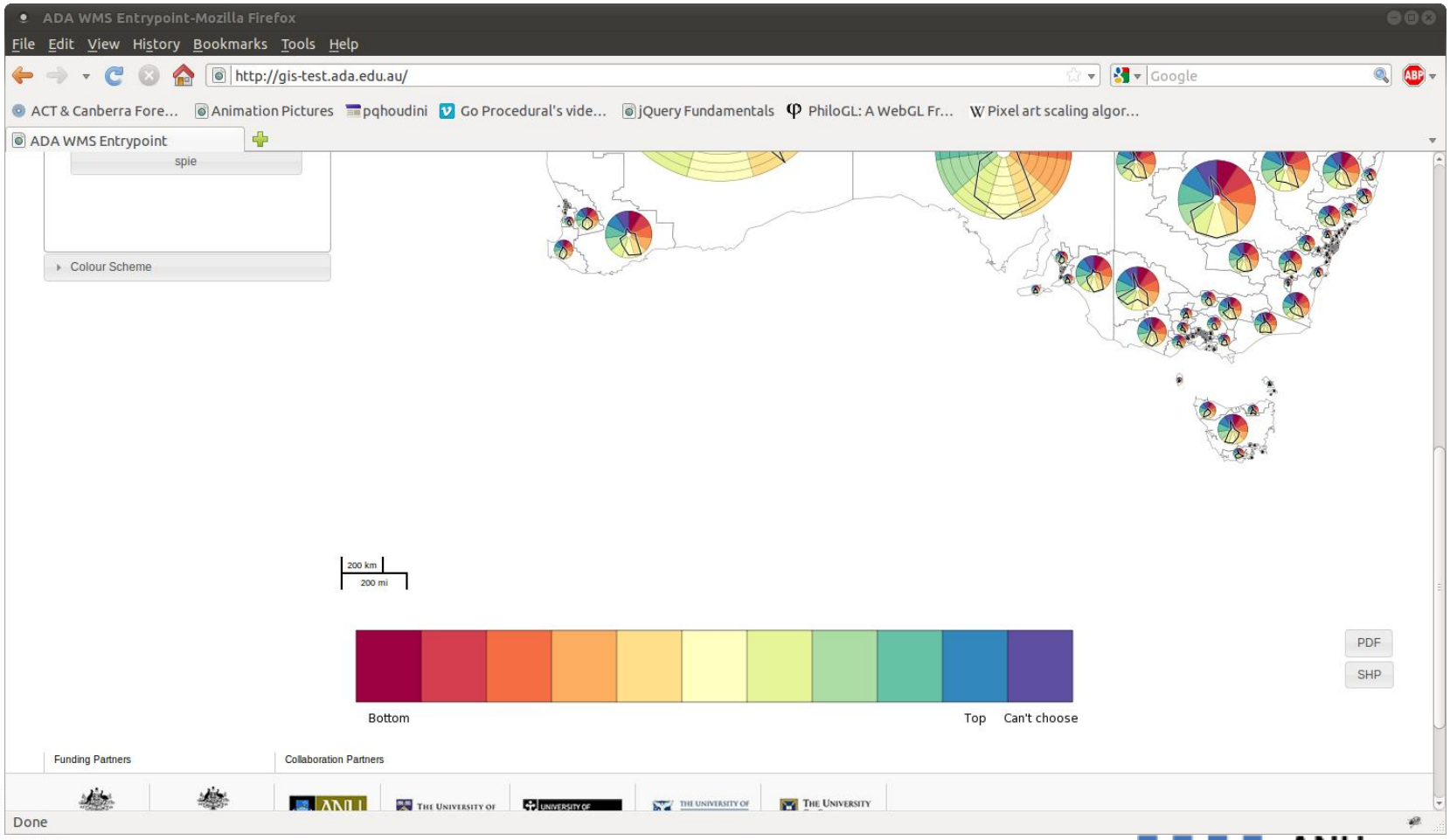
Ordinal variables – SPIE charts



Ordinal variables – legend



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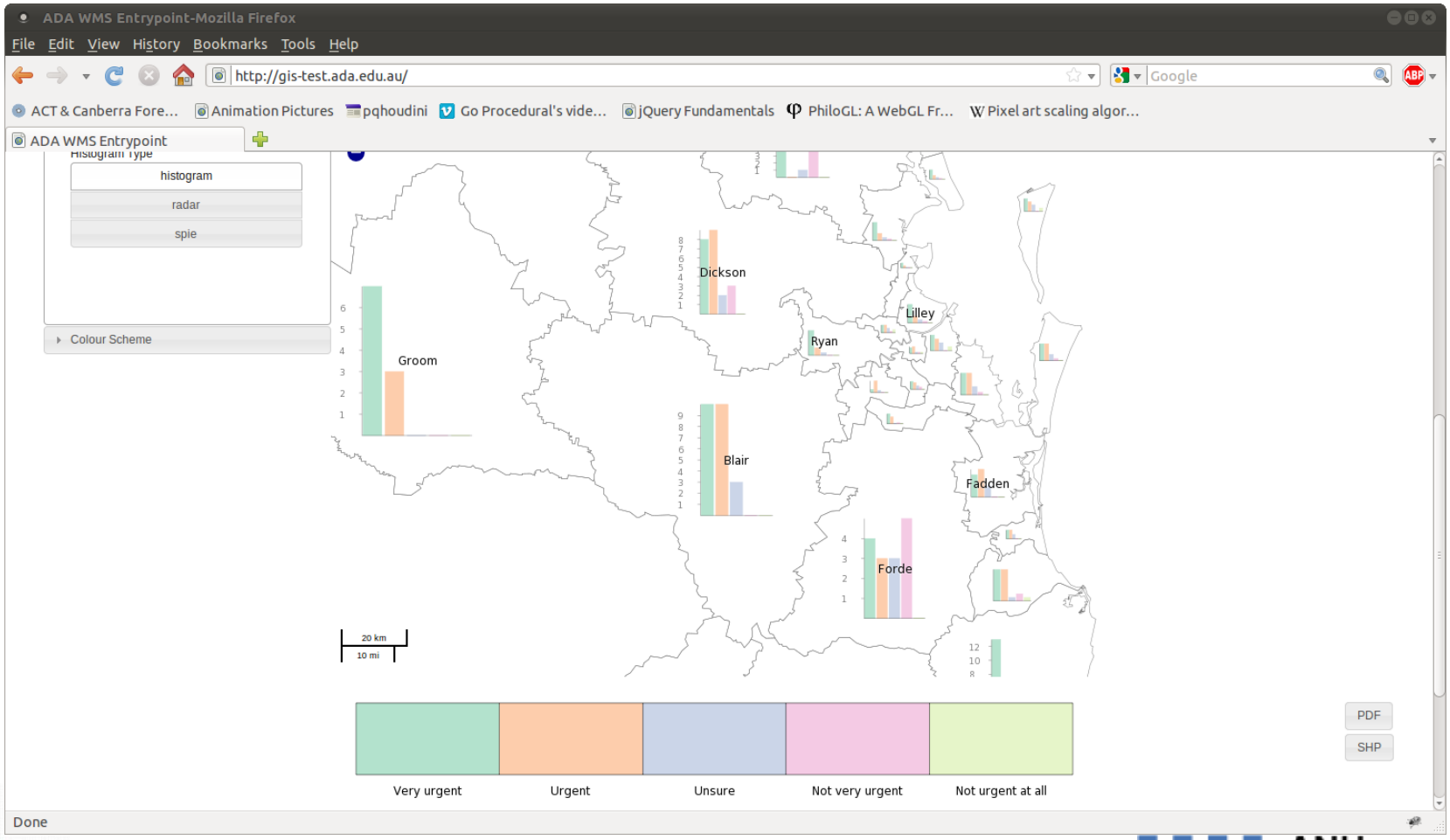
Nominal variables - histogram



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Data export

Currently two types of “export”

- Image PDF
- ESRI shape file (*.shp)

- Original unit record data file is available through standard means (Nesstar download)

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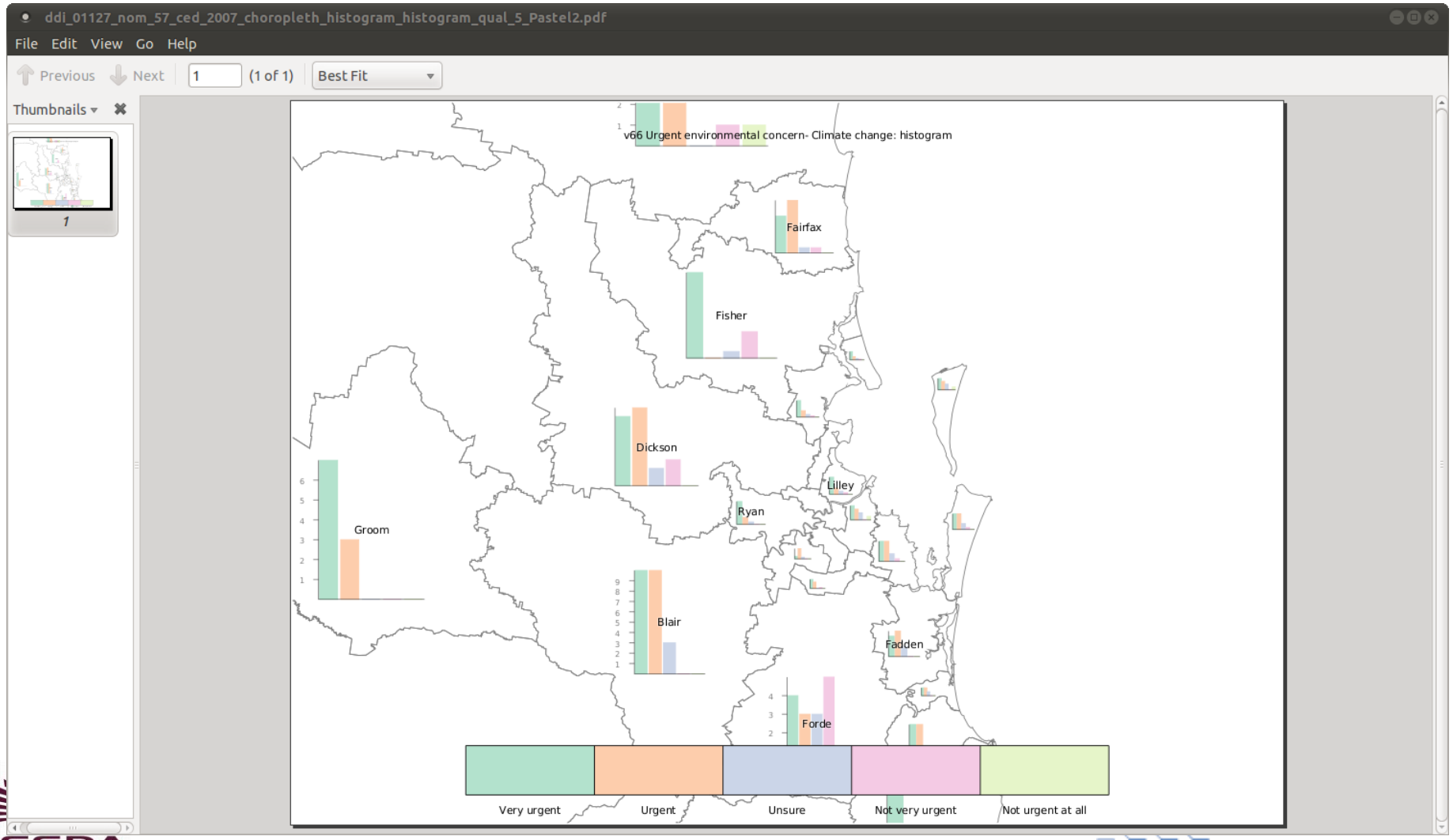
PDF Export



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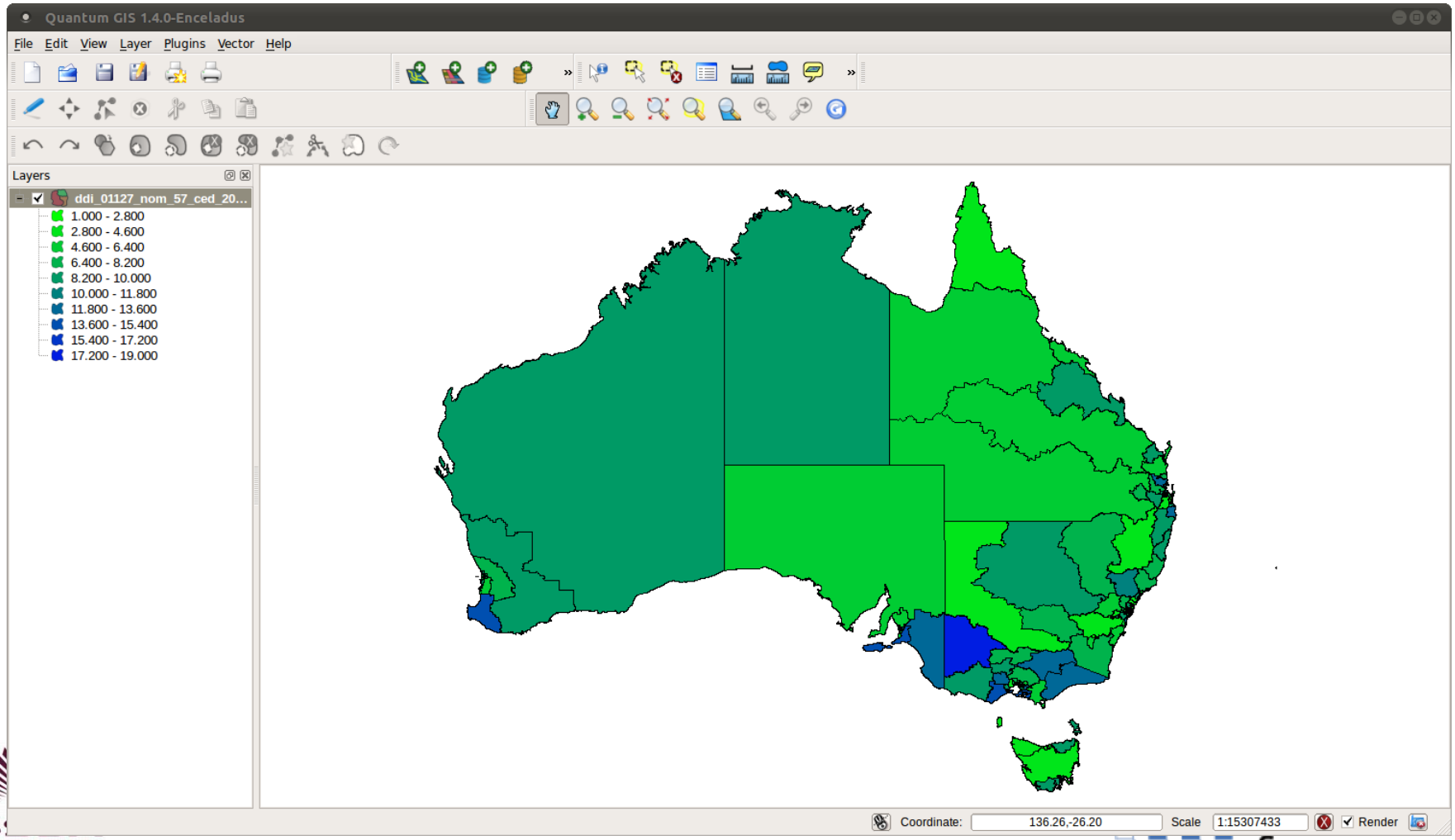
Shape file export (Imported to QGIS)



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Current capabilities

- Javascript-based menu, OpenLayers map

Options:

- Map types by variable type (nom/ord/int)
- Percentages, modes and histograms
- Standard histograms, radars and spie charts
- Variable colour schemes
- Chloropleth and dorling layouts
- Data export

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Future directions

- Online geospatial analysis: current tool is univariate/bivariate only
- Integration across datasets (mash-up capability)
- Extending to broader range of datasets
- Spatio-temporal analysis (incorporating longitudinal visualisation – parallel ADA project through VizLab)
- Historical spatial boundaries (moving forwards by going backwards)
- Open GIS formats for data dissemination

- Questions or comments?

For further information

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